GG-W-259a
JULY 22, 1959
SUPERSEDING
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May 24, 1948

FEDERAL SPECIFICATION

WHEEL, BUFFING, CLOTH, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—This specification covers cloth wheels suitable for mounting on tapered lathe chucks and for buffing dental appliances.

1.2 **Classification.**

1.2.1 *Classes.*—Wheels shall be of the following classes, as specified (see 6.1):

Class 1—30 ply.
Class 2—42 ply.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CCC-T-191—Textile Test Methods.
PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-B-10377—Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL-L-10547 — Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific

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procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Muslin.*—The cloth shall be a high quality, fine, unbleached muslin having 80 yarns per inch in the warp, and 92 yarns per inch in the filling, and weighing 4.1 ounces per square yard. The muslin shall meet the requirements specified herein, when tested in accordance with 4.4.1 and 4.4.2. A tolerance of two yarns per inch for warp and filling, and 3 percent for weight per square yard shall be allowed.

3.1.2 *Thread.*—The thread used for stitching shall be soft finish, unbleached thread with a minimum breaking strength of 6.0 pounds, when tested as specified in 4.4.3.

3.2 **Construction.**—Wheel shall be made of plies of muslin stitched together with the centers hardened. The wheel center shall be punched with a hole not less than ⅛ inch nor more than 3/16 inch in diameter for mounting on tapered lathe chucks. The wheel shall be flat across its face. The wheels shall be prefaced by machine to make it ready for immediate mounting on dental lathe chuck. The nap on the surface of the wheel shall be not less than 1/16 inch in depth. Wheels shall conform to the requirements shown in table I for the class specified.

3.2.1 *Plies.*—The plies of muslin shall be of uniform thickness and texture. They shall extend to the outer edge of the wheel and shall be turned one-twelfth to one-eighth of a circle so that the warp yarns of one ply are turned 30 to 45 degrees to the warp yarns of the group above and below it. The number of plies in each wheel shall be as shown in table I, for the class specified.

3.2.2 *Stitching.*—Stitching shall be concentric machine sewing and shall conform to the stitching requirements of table I, for the class specified. There shall be a minimum of 4½ stitches and a maximum of 7 stitches to an inch.

3.2.3 *Center hardening.*—The centers of wheels shall be hardened with a suitable hardening agent that will not deteriorate and will prevent the wheel from riding up on the lathe spindle.

3.3 **Workmanship.**—Workmanship shall be first class throughout. The muslin used shall be free from holes, mends, sewed pieces, or other defects. The finished buffing wheels shall be free from any other defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier responsibilities for inspection.**—Such examinations and tests as are set forth in this specification or as shall other-

TABLE I.—*Construction*

Class	Diameter of wheel (tolerance ± 1/16 inch)	Plies (tolerance plus 3)	Rows of stitching	Diameter of innermost row of stitches (tolerance ± 1/16 inch)	Distance between outside row of stitches to peripheral edge of wheel ± ⅛ inch)
	Inches			*Inch*	*Inch*
1	4	30	4	9/16	¾
2	4	42	4	9/16	¾

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wise be appropriate or necessary to insure that supplies conform to specification requirements shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing, (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 *For examination.*—Sampling for examination shall be in accordance with instructions furnished by the procuring activity.

4.2.2 *For component testing.*—Sufficient samples of the components shall be selected from each lot of raw materials to perform tests as required by tests specified herein. The sample size for each test shall be as indicated for inspection level L2 in table VII in appendix to Military Standard MIL–STD–105. No test result shall fall below the minimum or exceed the maximum values specified. The sample unit for testing shall be the quantity necessary to perform all required tests one time each. Not less than three samples shall be taken regardless of lot size. Lot sizes are expressed as follows:

Cloth	yards
Thread	cone or spool

4.2.3 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination.—The wheels shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests.—Tests of the component materials used in fabrication of the wheels shall be made as specified herein. Other tests may be made, if necessary, to determine that the wheels comply with this specification.

4.4.1 *Weight per square yard.*—Test for weight per square yard shall be conducted in accordance with method 5041 of Federal Specification CCC–T–191.

4.4.2 *Thread count.*—Test for thread count shall be conducted in accordance with method 5050 of Federal Specification CCC–T–191.

4.4.3 *Breaking strength.*—Test for breaking strength shall be conducted in accordance with method 4100 of Federal Specification CCC–T–191.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Unit of issue.*—One wheel, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table II shall be packaged in each unit, intermediate and exterior container, as applicable for the required level of protection specified in procurement document.

5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to

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TABLE II.—*Packaging quantities*

Federal Stock Number	Unit package	Intermediate package	Exterior container
6520–568–4150	1 unit	12 units	360 units

be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a bag fabricated from virgin polyethylene seamless tubing having an average wall thickness of not less than 0.002 inch with no wall thickness at any point below 0.00125 inch. Excess air shall be expelled or evacuated, and closure effected by heat sealing.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking required by the contract or order, marking shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Unit packages.*—Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor. The required information shall be

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printed either directly on the bag or on a label inserted within the bag in such manner as to permit ready identification.

5.3.2.2 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2.3 *Exterior container.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129, except that paragraphs 6.2(c), 6.2(d), 6.5, and 7.5.1 shall not be applicable. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Class required (see 1.2).
(*c*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item appearing in the Armed Services Medical Stock List:

6520–568–4150 WHEEL, BUFFING, Cloth, 4 inch.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 Transportation description.—Transportation description and minimum weight application to this commodity are:

Rail:
Wheels, buffing or polishing, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:
Wheels, buffing or polishing, not otherwise indexed.
Motor volume minimum weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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